Experiment No.	6
Experiment Title	Bagging, Boosting, and Stacked Ensemble Models [1]
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1 Objective

- To understand ensemble learning strategies: Bagging, Boosting, and Stacking.
- To implement Bagging and Boosting classifiers.
- To build a Stacked Ensemble model using multiple base learners.
- To compare ensemble models in terms of accuracy, stability, and generalization.
- To analyze the effect of ensemble methods on bias and variance.

2 Problem Statement

Breast cancer diagnosis based on cell-nuclei measurements from a digitized image of a fine needle aspirate (FNA) of a breast mass is a binary classification problem. Given 30 numeric features describing the size, texture, and shape of the cell nuclei, the goal is to predict whether a tumour is **Malignant (M)** or **Benign (B)**. Three ensemble learning strategies – Bagging, Boosting, and Stacking – are implemented, tuned using 5-fold cross-validation, and compared on standard classification metrics to study how each strategy affects bias, variance, and generalization.

3 Dataset Description

Dataset Name	Wisconsin Diagnostic Breast Cancer (WDBC) Dataset
Dataset Source	UCI Machine Learning Repository – https://archive.ics.uci.edu (accessed via the identical, bundled copy in `sklearn.datasets.load_breast_cancer`)
Number of Samples	569
Number of Features	30 numerical attributes (mean, standard error, and "worst" values of 10 base measurements per cell nucleus)
Number of Classes	2 (Malignant = 212 samples, Benign = 357 samples)
Missing Values	0
Train-Test Split	80%–20% stratified split (455 training / 114 testing samples), 5-fold stratified cross-validation used for hyperparameter tuning

4 Exploratory Data Analysis

The following visualizations were used to understand the dataset structure, class balance and relationships among features.

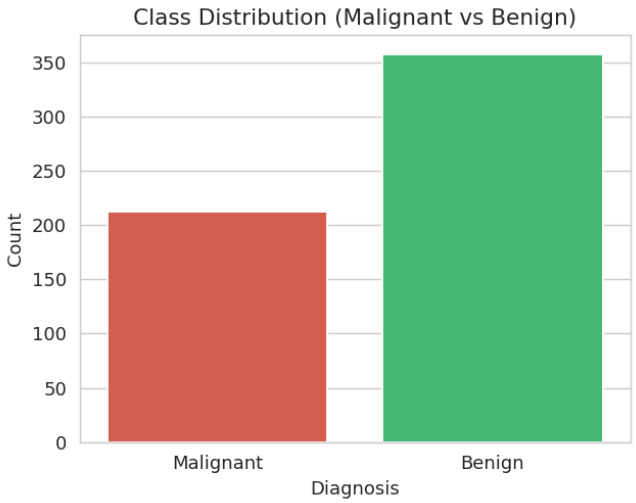


Figure 1: Class distribution of the WDBC dataset (Malignant vs Benign).

Inference: The dataset is moderately imbalanced, with 357 Benign and 212 Malignant samples (approximately 63:37). This imbalance is mild enough that accuracy remains informative, but precision, recall, and F1-score are also tracked to ensure the minority (Malignant) class is not overlooked. Stratified splitting and stratified cross-validation were therefore used throughout to preserve this class ratio in every fold.

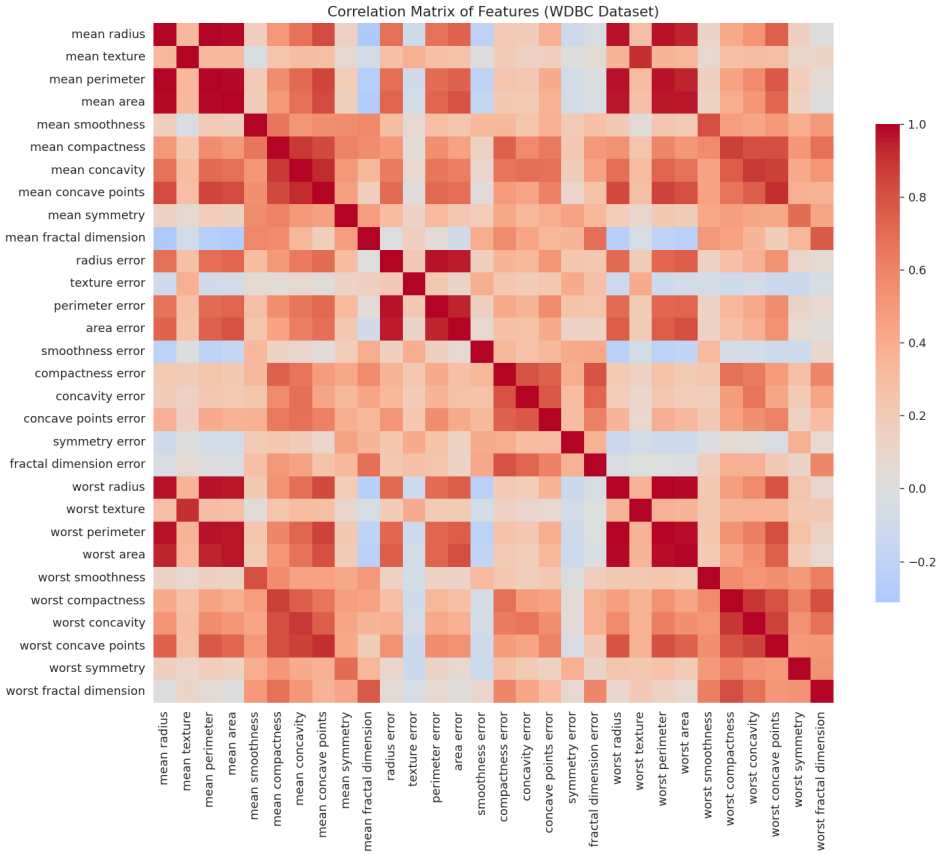


Figure 2: Correlation matrix of the 30 numerical features.

Inference: Many features are highly correlated, particularly the "mean", "standard error" and "worst" variants of the same base measurement (e.g., radius, perimeter, and area), since these are mathematically related quantities. This multicollinearity suggests that tree-based ensembles (which are largely insensitive to correlated features) are well suited to this dataset, whereas linear models may benefit from the diversity introduced by stacking.

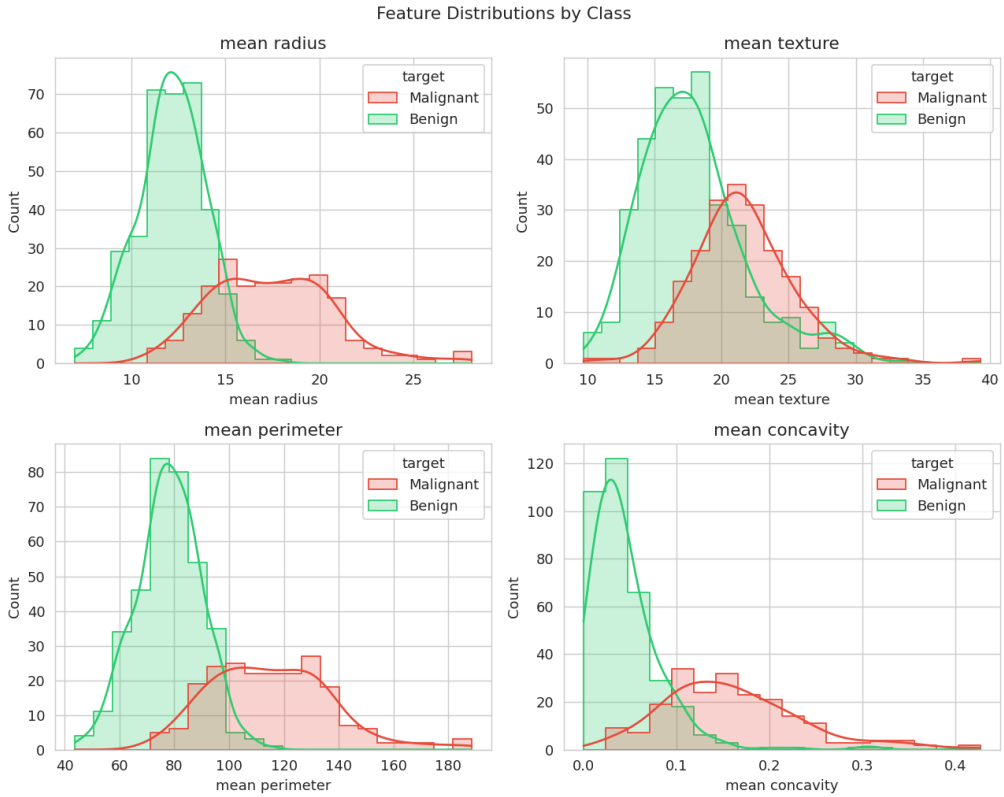


Figure 3: Distribution of selected features (mean radius, texture, perimeter, concavity) by class.

Inference: Malignant samples consistently show higher values of mean radius, perimeter, and concavity compared to benign samples, with visible separation between the two class distributions. This confirms that these morphological features carry strong discriminative signal, which explains why relatively simple base learners (Decision Trees, Naive Bayes) already perform well, and why ensembling further sharpens the decision boundary.

5 Data Preprocessing

- **Missing value handling:** The dataset contains no missing values, so no imputation was required.
- **Encoding:** The target variable is already encoded as binary (0 = malignant, 1 = benign) by scikit-learn; no additional encoding was needed since all 30 features are numeric.
- **Feature scaling:** All 30 features were standardized using `StandardScaler` (zero mean, unit variance) since Bagging, Boosting, SVM, and Logistic Regression base/meta learners are all sensitive to feature scale (distance- and gradient-based).
- **Train-test split:** An 80–20 stratified split was used (455 train / 114 test samples), preserving the malignant/benign ratio in both sets.
- **Feature engineering:** No manual feature engineering was performed; the 30 provided features were used directly, as they already encode informative geometric

statistics of the cell nuclei.

6 Machine Learning Algorithm

Bagging (Bootstrap Aggregation)

Working principle: Bagging trains B base learners independently, each on a different bootstrap sample (random sample with replacement) of the training data, optionally also sub-sampling features. Predictions are aggregated by majority vote (classification).

Mathematical formulation: Given training set D of size n, B bootstrap samples $D_1, \ldots, D_B$ are drawn (each of size n, sampled with replacement). A base model $h_b(x)$ is trained on each D_b, and the ensemble prediction is

$$\hat{y}(x) = \text{mode}\{h_1(x), h_2(x), \ldots, h_B(x)\}.$$

Because the h_b are trained on different resamples, averaging/voting reduces the *variance* of the ensemble relative to a single high-variance learner, since $\text{Var}(\bar{h}) \approx \text{Var}(h)/B$ under (approximate) independence of the h_b.

Advantages: Reduces variance, resistant to overfitting for high-variance base learners (e.g., unpruned decision trees), parallelizable since models are trained independently.

Limitations: Does not reduce bias – if the base learner is systematically biased, bagging will not correct it; less interpretable than a single tree; can be computationally expensive with many/large base learners.

Boosting (AdaBoost / Gradient Boosting)

Working principle: Boosting builds an additive ensemble sequentially: each new base learner focuses on the mistakes (misclassified or high-residual samples) of the ensemble built so far. AdaBoost re-weights misclassified samples; Gradient Boosting fits new learners to the negative gradient (residuals) of the loss function.

Mathematical formulation (AdaBoost): At iteration m, sample weights w_i are updated based on classification error ϵ_m, and the learner's weight is $\alpha_m = \frac{1}{2}\ln\left(\frac{1-\epsilon_m}{\epsilon_m}\right)$. The final classifier is $H(x) = \text{sign}\left(\sum_{m=1}^{M} \alpha_m h_m(x)\right)$.

Mathematical formulation (Gradient Boosting): The ensemble is built as $F_m(x) = F_{m-1}(x) + \eta \cdot h_m(x)$, where h_m is trained to approximate the negative gradient of the loss $L(y, F_{m-1}(x))$ and η is the learning rate.

Advantages: Reduces bias, often achieves very high accuracy, can handle diverse loss functions.

Limitations: Sequential training (harder to parallelize), more sensitive to noisy data and outliers (especially AdaBoost), higher risk of overfitting if the number of estimators/learning rate is not tuned carefully.

Stacked Ensemble (Stacking)

Working principle: Stacking trains multiple heterogeneous base learners on the training data, then trains a meta-learner on the (out-of-fold) predictions of the base learners to learn the optimal way to combine them.

Mathematical formulation: Given base learners $h_1, \ldots, h_K$, out-of-fold predictions $p_k(x) = h_k(x)$ are used as meta-features. The meta-learner g (here, Logistic Regression) is trained as $\hat{y}(x) = g\big(p_1(x), p_2(x), \ldots, p_K(x)\big)$.

Advantages: Can outperform any individual base model by exploiting complementary strengths of diverse learners; the meta-learner learns non-trivial combination weights rather than simple averaging/voting.

Limitations: More complex and computationally expensive (requires cross-validated base-model predictions); higher risk of overfitting the meta-learner if not properly cross-validated; less interpretable than a single model.

7 Experimental Setup

Python Version	3.12.3
Scikit-Learn Version	1.8.0
IDE	Jupyter Notebook
Operating System	Windows
Hardware	Hardware LAPTOP-QOD0ENML

8 Hyperparameter Selection

Parameter	Value
Bagging: n_estimators	50
Bagging: max_samples	0.5
Bagging: max_features	0.5
Boosting (AdaBoost): n_estimators	100
Boosting (AdaBoost): learning_rate	1.0
Stacking: SVM C	1
Stacking: final_estimator (LogisticRegression) C	10

9 Results

Metric	Value (Stacked Ensemble, best model)
Accuracy	97.37%
Precision	0.9859
Recall	0.9722
F1-score	0.9790
ROC-AUC	0.9947

Hyperparameter Evaluation Results

Table 2: Bagging Hyperparameter Evaluation (Top results, 5-fold CV)

n_estimators	max_samples	Avg CV Accuracy (%)	Avg CV F1 Score
50	0.5	96.04	0.9681
100	0.8	96.04	0.9683
100	1.0	96.04	0.9684
50	1.0	96.04	0.9683
10	1.0	95.60	0.9648

Table 3: Boosting Hyperparameter Evaluation (Top results, 5-fold CV)

Model	n_estimators	learning_rate	Avg CV Accuracy (%)	Avg CV F1 Score
AdaBoost	100	1.0	98.02	0.9844
AdaBoost	150	1.0	97.80	0.9827
GradientBoosting	150	0.2	97.36	0.9789
GradientBoosting	100	0.2	96.92	0.9755
AdaBoost	50	1.0	96.70	0.9740

Table 4: Stacked Ensemble Evaluation (Top results, 5-fold CV)

SVM C	Final Estimator C	Avg CV Accuracy (%)	Avg CV F1 Score
1	10	97.14	0.9771
10	1	96.92	0.9754
1	1	96.70	0.9737
10	10	96.70	0.9737
1	0.1	96.48	0.9720

Base Models: SVM, Naïve Bayes, Decision Tree Meta Learner: Logistic Regression

Performance Comparison of Ensemble Models (Held-out Test Set)

Table 5: Performance Comparison of Ensemble Models

Model	Accuracy (%)	Precision	Recall	F1 Score
Bagging	95.61	0.9589	0.9722	0.9655
Boosting (AdaBoost)	95.61	0.9467	0.9861	0.9660
Stacked Ensemble	97.37	0.9859	0.9722	0.9790

10 Result Visualization

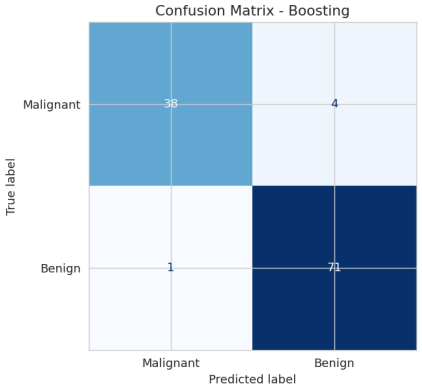


Figure 4: Confusion matrices on the test set: Bagging (left) and Boosting/AdaBoost (right).

Inference: Bagging misclassifies 3 malignant samples as benign and 2 benign samples as malignant, while Boosting (AdaBoost) misclassifies 4 malignant samples as benign but only 1 benign as malignant, giving it a higher recall on the malignant class at a slight cost to precision. Both models show that most classification errors occur near the decision boundary where morphological features of the two classes overlap.

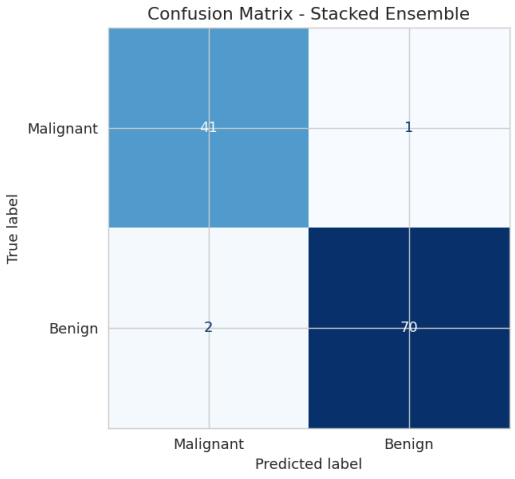


Figure 5: Confusion matrix on the test set: Stacked Ensemble.

Inference: The Stacked Ensemble achieves the fewest total misclassifications (3 out of 114 test samples), with only 1 benign sample misclassified as malignant and 2 malignant samples misclassified as benign. This confirms that combining diverse base learners (SVM, Naïve Bayes, Decision Tree) via a meta-learner captures complementary decision boundaries that individual ensemble strategies miss.

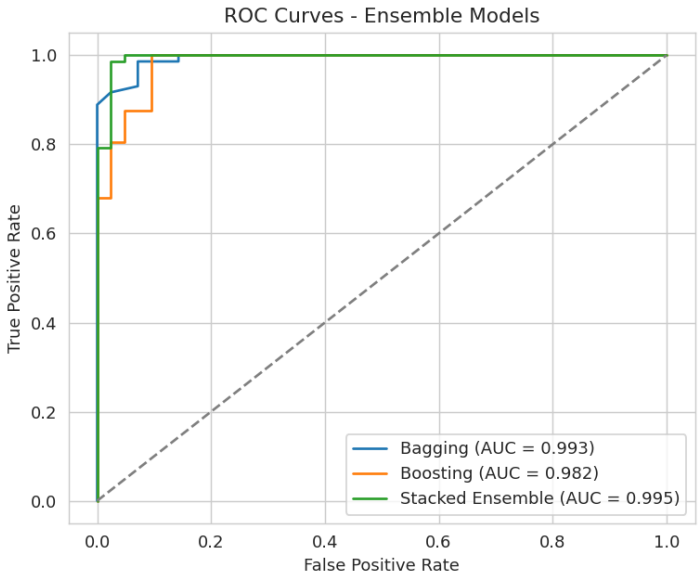


Figure 6: ROC curves for Bagging, Boosting, and Stacked Ensemble on the test set.

Inference: All three ensembles achieve very high ROC-AUC (>0.98), confirming excellent class separability. The Stacked Ensemble (AUC = 0.9947) and Bagging (AUC = 0.9931) obtain marginally higher AUC than Boosting (AUC = 0.9818), indicating slightly better-calibrated ranking of predicted probabilities across all classification thresholds.

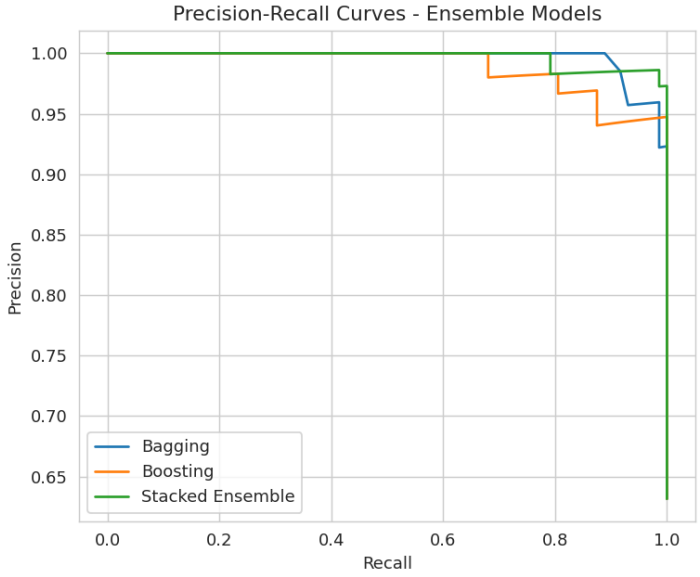


Figure 7: Precision-Recall curves for the three ensemble models.

Inference: The precision-recall curves remain close to the top-right corner (precision and recall near 1) for all three models across most of the recall range, reflecting the strong overall discriminative power of the feature set. The Stacked Ensemble maintains high precision even at high recall, making it the most reliable choice when false negatives (missed malignant cases) must be minimized without sacrificing precision.

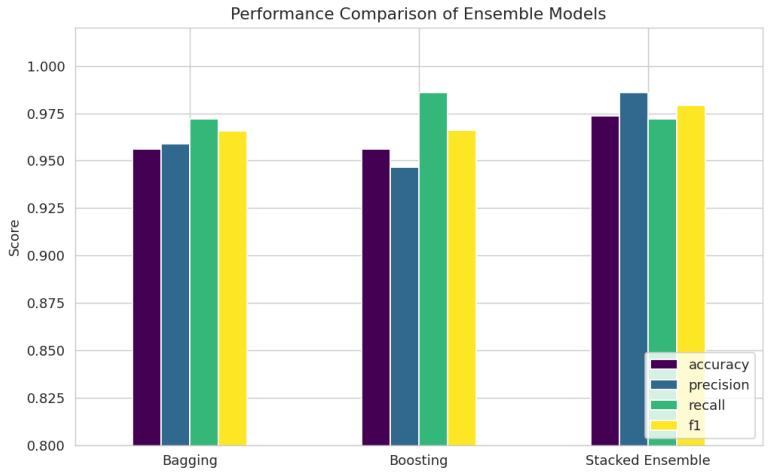


Figure 8: Bar chart comparison of Accuracy, Precision, Recall, and F1-score across models.

Inference: The Stacked Ensemble leads on accuracy, precision, and F1-score, while Boosting attains the highest recall. Bagging is competitive on all metrics without being the top performer on any single one, reflecting its role as a variance-reduction technique rather than a bias-correction or combination technique.

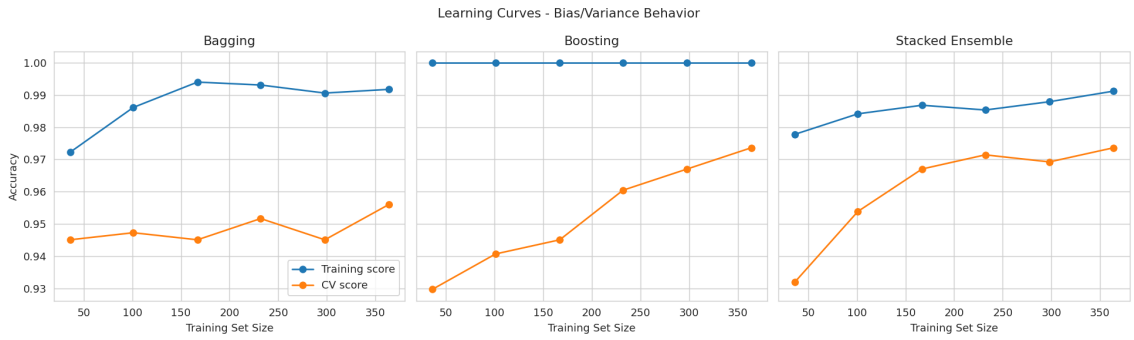


Figure 9: Learning curves (training vs cross-validation accuracy) for the three ensemble models.

Inference: Bagging shows a small, stable gap between training and CV accuracy across all training sizes, consistent with its variance-reduction property. Boosting's training accuracy stays near 1.0 while CV accuracy converges upward with more data, indicating it fits the training data very closely (lower bias) but requires care to avoid overfitting. The Stacked Ensemble shows the narrowest, most consistent gap at larger training sizes, suggesting the best bias-variance trade-off of the three.

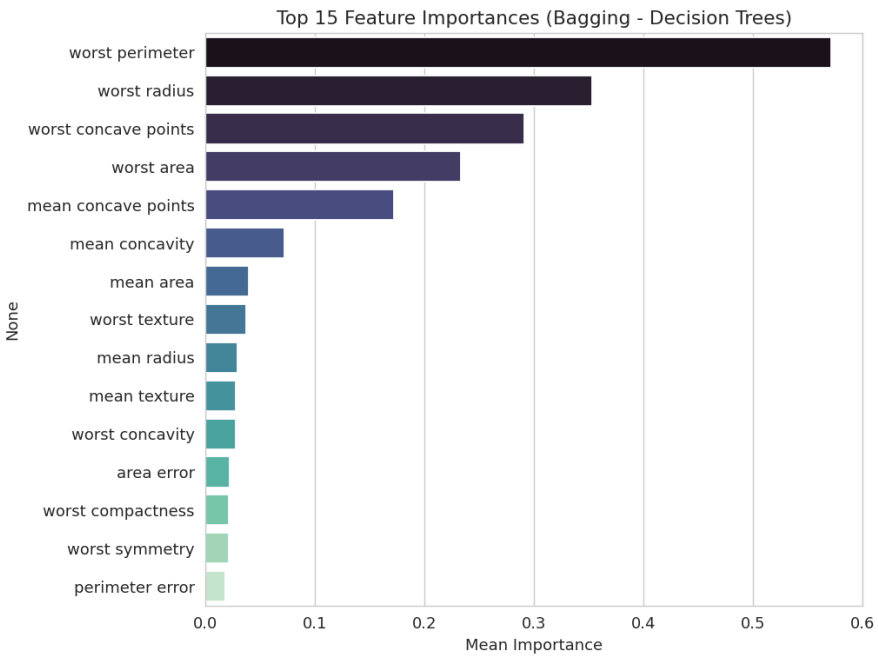


Figure 10: Top 15 feature importances averaged across the Bagging ensemble's Decision Trees.

Inference: Features such as "worst perimeter", "worst area", "worst concave points", and "mean concave points" dominate the ensemble's decisions, consistent with the class separation observed during EDA. This agreement between the unsupervised (correlation/ distribution) analysis and the supervised feature-importance ranking increases confidence that the model is learning genuine, clinically-relevant morphological signal rather than noise.

11 Discussion

Across all three ensemble strategies, test accuracy exceeded 95%, confirming that the WDBC features are highly discriminative for this task. Bagging effectively reduced the variance of an otherwise unstable single Decision Tree, producing consistent performance with a low training-CV gap. Boosting (AdaBoost outperforming Gradient Boosting on this dataset) achieved the highest recall, an important property in a medical diagnosis context where false negatives (missed malignant tumours) are more costly than false positives. The Stacked Ensemble, combining SVM, Naïve Bayes, and Decision Tree base learners through a Logistic Regression meta-learner, achieved the best overall balance of accuracy, precision, and F1-score, demonstrating the value of heterogeneous base learners whose errors are less correlated than those of the homogeneous Bagging/Boosting ensembles. A limitation of this study is the moderate dataset size (569 samples), which increases the variance of cross-validated estimates; a larger dataset or repeated cross-validation would improve confidence in the ranking of the models. Future improvements could include exploring additional/varied base learners for stacking, cost-sensitive learning to further prioritize recall on the malignant class, and more exhaustive hyperparameter search (e.g., randomized or Bayesian search) for Boosting and Stacking.

12 Conclusion

Bagging, Boosting, and Stacked Ensemble models were implemented and evaluated on the Wisconsin Diagnostic Breast Cancer dataset using 5-fold cross-validated hyperparameter tuning. All three ensembles achieved strong test performance (95.6%–97.4% accuracy), with the **Stacked Ensemble** performing best overall (97.37% accuracy, F1 = 0.979, ROC-AUC = 0.9947), followed closely by Bagging and Boosting. The experiment demonstrates concretely how Bagging reduces variance, Boosting reduces bias and maximizes recall, and Stacking leverages diverse base learners to achieve superior generalization compared to any individual ensemble strategy.

13 References

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[1 Scikit-learn Developers, "Ensemble methods," scikit-learn documentation. [Online]. Available: https://scikit-learn.org/stable/modules/ensemble.html

[2 Scikit-learn Developers, "sklearn.ensemble.BaggingClassifier," scikit-learn documentation. [Online]. Available: https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.BaggingClassifier.html

[3 W. Wolberg, O. Mangasarian, N. Street, and W. Street, "Breast Cancer Wisconsin (Diagnostic) Data Set," UCI Machine Learning Repository, 1995. [Online]. Available: https://archive.ics.uci.edu